[D.3.2] Universal AI: AIXI

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Iliad Intensive

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The big picture: one agent, two halves

- **Universal AI** asks: what would an *optimally intelligent* agent look like, given unlimited compute and the weakest possible assumptions?
- It splits into two problems:
  1. **Prediction (Solomonoff):** passively observe a sequence $x_1, x_2, \ldots$ and predict the next symbol. *How should you learn?*
  2. **Action (AIXI):** interact with an unknown environment, take actions, collect reward. *How should you act?*
- **The common engine:** a single **Bayesian mixture** ξ over a class $\mathcal{M}$ of candidate environments, weighted by a prior w_ν.
  - Solomonoff: ξ (eventually) predicts as well as the truth μ.
  - AIXI: the Bayes-optimal policy π^*_ξ (eventually) acts as well as the optimal policy π^*_μ for true environment μ.
- **Universal** choice: $\mathcal{M} =$ all computable environments, prior $w_\nu = 2^{-K(\nu)}$ (K is the "complexity" of the environment). Assumption "$\mu \in \mathcal{M}$" becomes "*the universe is computable*".

The Bayesian mixture ξ

Definition

Class $\mathcal{M} = \{\nu_1, \nu_2, \dots\}$ with prior weights $w_\nu > 0$, $\sum_\nu w_\nu = 1$.

$$\xi(x_{1:t}) := \sum_{\nu \in \mathcal{M}} w_\nu \, \nu(x_{1:t}).$$

- **Key fact (mixture is a predictor):** its one-step prediction is a *posterior-weighted* average of the members' predictions (you prove this today!)

$$\xi(x_t \mid x_{<t}) = \sum_{\nu \in \mathcal{M}} w(\nu \mid x_{<t}) \, \nu(x_t \mid x_{<t}), \qquad w(\nu \mid x_{<t}) = w_\nu \, \frac{\nu(x_{<t})}{\xi(x_{<t})}.$$

- **Posterior update is multiplicative:** $w(\nu \mid x_{1:t}) = w(\nu \mid x_{<t}) \cdot \dfrac{\nu(x_t \mid x_{<t})}{\xi(x_t \mid x_{<t})}$: we reweight by how well ν predicted vs. ξ (Bayes' rule!)

From prediction to action

- Now the agent doesn't just watch: it **acts**, and its actions shape what it sees.
- At each step t: the agent picks action a_t, the environment returns a **percept** $e_t = (o_t, r_t) =$ observation + reward.

$$h = a_1\, e_1\, a_2\, e_2\, a_3\, e_3\, \cdots \qquad \text{(history } æ_{<t}\text{)}$$

- Spaces: actions $\mathcal{A}$, observations $\mathcal{O}$, rewards $\mathcal{R} \subset [0,1]$, percepts $\mathcal{E} = \mathcal{O} \times \mathcal{R}$, and (finite) histories $\mathcal{H}^* := (\mathcal{A} \times \mathcal{E})^* := \cup_{t=0}^{\infty} (\mathcal{A} \times \mathcal{E})^t$.
- **No Markov / MDP assumption!** Both the policy and environment can depend on all prior percepts.

The three players

Policy π

$\pi(a_t \mid æ_{<t}) : \mathcal{H} \to \Delta\mathcal{A}$ is the agent's distribution over the next action given history.

Environment ν

$\nu(e_t \mid æ_{<t}\, a_t) : \mathcal{H} \times \mathcal{A} \to \Delta\mathcal{E}$ is the environment's distribution over the next percept given history and action. True (unknown) environment is μ.

Interaction measure ν^π

Run π inside ν and multiply along the history:

$$\nu^\pi(æ_{t:m} \mid æ_{<t}) := \prod_{k=t}^{m} \pi(a_k \mid æ_{<k})\, \nu(e_k \mid æ_{<k} a_k).$$

Value, optimality, and AIXI

Value function (assuming geometric discount γ)

$$V_\nu^{\pi,m}(æ_{<t}) = (1-\gamma)\,\mathbb{E}_\nu^\pi[G_{t-1:m} \mid æ_{<t}] \text{ with } G_{t-1:m} = \sum_{k=t}^{m} \gamma^{k-t} r_k$$

$$= (1-\gamma) \sum_{æ_{t:m}} \nu^\pi(æ_{t:m} \mid æ_{<t}) \sum_{k=t}^{m} \gamma^{k-t} r_k$$

The **infinite-horizon value** is defined as the pointwise limit

$$V_\nu^\pi \equiv V_\nu^{\pi,\infty}(æ_{<t}) := \lim_{m\to\infty} V_\nu^{\pi,m}(æ_{<t}).$$

The **optimal value** is $V_\nu^{*,m}(æ_{<t}) := \sup_\pi V_\nu^{\pi,m}(æ_{<t})$, with $V_\nu^* \equiv V_\nu^{*,\infty}$.

- **Optimal policy** π_ν^* satisfies $V_\nu^{\pi_\nu^*} = V_\nu^*$.
- **Bayes-optimal policy** $\pi_\xi^* \in \arg\max_\pi V_\xi^\pi$: acts optimally w.r.t. the *mixture* ξ.

AIXI

AIXI

π^*_ξ with $\mathcal{M}$ = all ≈computable environments[a] and prior $w_\nu = 2^{-K(\nu)}$. Write ξ_U for ξ with this choice of $\mathcal{M}$ and w_ν, so AIXI = $\pi^*_{\xi_U}$.

[a] lower-semicomputable semi-measures

Intuition: *Act optimally according to environments that you believe to be in, assuming the simplest environments consistent with observations are more likely to be true.*

AIXI, explicitly

max over actions, $\sum$ over percepts, choosing actions that maximise the ξ-weighted discounted return:

$$a_t = \arg\max_{a_t} \lim_{m\to\infty} \sum_{e_t} \max_{a_{t+1}} \sum_{e_{t+1}} \cdots \max_{a_m} \sum_{e_m} \xi(e_{t:m} \mid æ_{<t}\, a_{t:m}) \sum_{k=t}^{m} \gamma^{k-t} r_k$$

Result 1: On-policy value convergence

Theorem

For any $\mu \in \mathcal{M}$ and any policy π,

$$V_\xi^\pi(æ_{<t}) - V_\mu^\pi(æ_{<t}) \to 0 \qquad \mu^\pi\text{-almost surely.}$$

Intuition: *Following policy π, pretending the environment is ξ, eventually will act as if we knew the true environment μ.*

Result 2: AIXI can't be fooled

Theorem (deterministic μ)

If $V_\mu^*(æ_{<t}) > \varepsilon$ along history $æ_{<t}$, then

$$V_\xi^*(æ_{<t}) \;\geq\; w_\mu\,\varepsilon \;>\; 0.$$

Intuition: *Whenever good behaviour in μ is achievable, the Bayes-optimal agent secures at least a fraction proportional to the prior weight of μ.*

Result 3: Self-optimizing property (stretch goal)

Theorem

Let $\mathcal{M}$ be finite and fix a historic policy π generating the actions $a_{<t}$. If *some* policy can learn to act optimally in every $\nu \in \mathcal{M}$, then π^*_ξ does too: for all $\mu \in \mathcal{M}$,

$$V^*_\mu(æ_{<t}) - V^{\pi^*_\xi}_\mu(æ_{<t}) \to 0 \quad \mu^\pi\text{-a.s.}$$

Intuition: *If the class is learnable at all, the Bayes-optimal agent learns to act as well as the agent that knows the true μ.*

- **But not for AIXI:** the theorem needs $\mathcal{M}$ finite. Over all computable environments AIXI explores too little and is *not* asymptotically optimal [Ors13, LH15], though some other policies are (see the safety slides).

Takeaways

- **Solomonoff** solves induction: mix over all computable hypotheses, weighted by simplicity. Total prediction error $\leq K(\mu) \ln 2$.
- **AIXI** lifts this to action: the Bayes-optimal policy over the same universal mixture. Its value estimates converge on-policy, and it can't be fooled; but it explores too little to be guaranteed to act optimally in the limit.
- **The catch:** both are *incomputable* (K is incomputable; $\mathcal{M}$ is intractable to enumerate, the expectimax is intractable). They are gold standards, not computable algorithms, and the prior's $2^{-K(\nu)}$ hides a dependency on the choice of universal Turing machine: a bad choice of UTM can make AIXI behave arbitrarily badly [LH15].
- **Approximations:** MC-AIXI-CTW [VNH+11] gives an approximation using Context Tree Weighting to replace ξ and Monte-Carlo-Tree-Search to replace the expectimax.

Why care for safety? (I) What would an optimal agent do?

AIXI makes "what does a capable, reward-maximizing agent want?" a theorem-shaped question.

- **Reward hijacking.** Given a delusion box, a reward maximizer uses it [RO11]. An agent planning in an unknown world, unable to rule out that *the reward signal itself* is what it should optimize, is incentivized to intervene in the provision of reward [CHO22]; this argument underpins a *Science* policy paper on long-term planning agents [CKB+24].
- **Tampering & self-modification.** Causal influence diagrams show which designs remove the incentive to tamper with the reward function [EHKK21]; agents that score futures with their *current* utility preserve their goals [OR11, EFDH16].
- **Shutdown is fragile.** Reading the missing mass of a semimeasure as "death", merely *shifting* rewards turns AIXI from suicidal to self-preserving [MEH16]. Power-seeking also falls out as empowerment maximization [HT25].
- **Exploration vs. traps.** Asymptotic optimality is achievable (e.g. Thompson sampling [LLOH16]), but only by trying everything: an asymptotically optimal agent is eventually destroyed or incapacitated [CCH20].

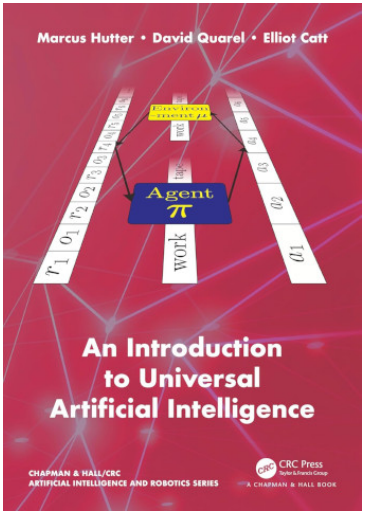
Why care for safety? (II) Mitigations and the bridge to LLMs

- **Myopia + boxing.** BoMAI maximizes reward only within an episode, in a sealed room; it is *asymptotically unambitious*, i.e. indifferent to seizing outside power [CVH20].
- **Pessimism + mentor.** Act on the *worst* plausible world models and defer to a mentor otherwise; the agent then rarely does anything unprecedented [CH20], and Golden Handcuffs (subjective reward range $[-L, 1]$) get sublinear regret vs. the mentor while no simple unsafe event is triggered by the agent before the mentor [EC26]. Same idea in conservative imitation [CHN22] and in Bayesian harm guardrails [BCM$^+$25] for Scientist AI [BCF$^+$25].
- **LLMs as approximate Solomonoff.** Transformers meta-trained on UTM outputs approximate the universal mixture [GMGH$^+$24]; prompting is then Bayesian conditioning, and some targets are unreachable by prompts alone [GWGM$^+$25]. Being KL-η close to ξ costs $O(\eta)$ on average but $\Theta(\eta/w_\mu)$ in environment μ: worst in rare, low-prior worlds [LT26].
- **Embedded agency.** AIXI sits outside its world and can't model itself [WH25]. Reflective oracles give a grain of truth for multi-agent play [WHLT25] and self-predicting universal agents [M$^+$25].

Want to work on this? AIXI Labs

- **AIXI Labs** ("AI X-risk Initiative"): an AI safety org founded in 2026 by Cole Wyeth, Aram Ebtekar, Michael Cohen, Matthias Dellago and Marcus Hutter [WEC+26].
- **Focus:** algorithmic information theory, continual RL, and AIXI. Stated agenda:
  - strengthen the *technical* case that ASI poses an existential risk;
  - prototype mitigations built on conservative AIXI variants (pessimism, corrigibility) and push frontier labs to adopt them;
  - port the AIXI analysis to LLM agents and test it empirically;
  - long-term deconfusion: learning in unrealizable environments, corrigibility.
- Funded by UK AISI's Alignment Project; runs a research fellowship and is hiring. For the worldview, read "The AIXI perspective on AI Safety" [Wye26].

Further reading: the textbook



An Introduction to Universal Artificial Intelligence
Hutter, Quarel & Catt, 2024 [HQC24]

- Today's material follows the book.
- Gentle intro, late-undergrad level.
- Covers prediction, agents, approximations and safety.
- Free PDF and lecture slides with exercises.

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